

MAKE A BUZZER ALARM!

The Midnight Mango Thief Adventure • Core Lesson Guide Book

PEDAGOGICAL ARCHITECTURE & BLUEPRINT

This primary STEM document converts the visual storyboard from the **Make a Buzzer Alarm! Slides** into a granular, page-by-page instructional script. The course introduces elementary hardware loop logic, switching concepts, and essential high-voltage boundaries to Grade 1 students through culturally resonant experiential narratives.

SLIDE 1 — COVER SLIDE

MAKE A BUZZER ALARM! THE MIDNIGHT MANGO THIEF

INTRODUCTORY PROTOCOL

Set a high-energy tone from the start. Introduce electrical engineering as an active superpower that helps solve immediate everyday problems around the house and garden.

SLIDE 2 — THE MISSION CONTEXT

THE GREAT MANGO ORCHARD HEIST

Slide Content: Story Mission: Ayaan's Alphonso mango is being nibbled every night! Can you help build a secret alarm to catch the thief? Help Ayaan protect his mangoes!

VOCAL DELIVERY SCRIPT: *"Namaste, young engineers! Welcome to today's big hardware challenge. Look closely at the screen! This is our friend Ayaan. His grandfather has grown a gorgeous, big, yellow Alphonso mango tree in their backyard orchard in Ratnagiri. But oh no! Every single midnight, a sneaky, mischievous little monkey climbs down the branches and takes a giant nibble out of the best fruits! Ayaan cannot stay awake all night long. Can we use our technology skills to build a secret electronic sound alarm trap to catch the thief in the act?"*

WHAT IS A CIRCUIT?

Slide Content: Imagine tiny 'Electricity Ants' marching through a wire path! They need a complete loop to deliver power and make things work.

- *Open Circuit:* The ants cannot jump across a broken wire — no sound!
- *Closed Circuit:* The ants march all the way to the buzzer and back — buzzer sings!

ANALOGY BREAKDOWN: "To make our alarm system work, we need to understand a secret path called a **CIRCUIT**. Let's imagine that inside every wire, there are billions of tiny, invisible, energetic 'Electricity Ants'! These ants love to run and carry energy. But they have one major rule: they will only start marching if they have a perfect, unbroken circular track to walk on! If there's a gap or a broken line, it's an **OPEN CIRCUIT**—the ants freeze because they can't jump! But if we connect the line perfectly, it becomes a **CLOSED CIRCUIT**. The ants march all the way through the sound maker and back home!"

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[BATTERY] ====(Ants Marching)===> [BUZZER] ====(Return Loop)===> [BATTERY]
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THE METRO & RICE LIGHTS EXAMPLE

Slide Content: Let's understand circuits with fun examples from India! When electricity ants have a complete path, everything works perfectly.

- *Metro Rail:* Full circle track means train moves nonstop. Broken track stops the train!
- *Diwali Lights:* One cut wire makes all the rice lights go off!

The City Metro Loop

Think of the busy Metro trains in Delhi, Mumbai, or Bangalore. If the circular track is entirely built and continuous, the train runs smoothly from station to station. If a section is removed for repairs, the loop breaks and the train stops instantly.

Festival Rice Lights

During festive celebrations like Diwali or Eid, families hang beautiful twinkling rice lights along the balconies. If just one tiny wire gets clipped or snapped by a window edge, the entire decorative sequence goes dark immediately because the ant pathway is open!

SLIDE 5 — ACTUATOR BREAKDOWN

MEET THE SOUND MAKER: THE BUZZER!

Slide Content: The buzzer is a magical part that turns the electricity ants' march into sound! When the ants complete their journey through the circuit, the buzzer sings!

- Doorbell goes 'Ding-Dong' when someone visits!
- School period bell rings to tell you it's time for the next class!
- Microwave beeps when your food is ready to eat!

INTERACTIVE PROMPTING: "Let's discover our alarm's mouth—the piezo buzzer! This little component takes the running speed of our electricity ants and turns it instantly into clear sound waves. Where have you heard this sound today? Yes! When a delivery person pushes your apartment doorbell button and it sings 'Ding-Dong'! Or when the school hallway bell loudly echoes across classrooms to tell you it's recess time!"

SLIDE 6 — CONTROL MECHANISM INTRODUCTION

THE PRESSURE SWITCH: YOUR ALARM'S TRIGGER

Slide Content: The pressure switch is the secret trigger for your alarm! When you push the pressure plate down, the electricity ants can finally march through and make the buzzer sing! Push the plate, ants can march through!

- Fridge door buzzer beeps when door opens too long.
- Welcome doormat bells ring when guests step on them!

How the Trigger Logic Functions:

Normally, the switch hangs out in an open state—holding the wires apart so the circuit stays silent. The moment the mango thief steps his foot down on our pressure plate, the mechanical weight forces the metal internal pads to touch, closing the gap instantly.

SLIDE 7 — INTERACTIVE ASSEMBLY STEP 1

STEP 1 - CONNECT BATTERY TO BUZZER

Slide Content: Drag the red wire from the 9V battery to the buzzer's red terminal. This connects the power source to your alarm! Watch carefully as the electricity ants get ready to march through your circuit. Wah! You connected the power!

UI GUIDE INTERACTION: "Alright, mini engineers, stretch your fingers out! Look at our digital workbench screen. We have a heavy-duty yellow 9V battery pack. Touch the positive red wire line, pull it gently across the screen, and snap it directly onto the red input leg of our buzzer. Outstanding! The power supply is locked into position."

 **WAH! YOU CONNECTED THE POWER!** 

SLIDE 8 — INTERACTIVE ASSEMBLY STEP 2

STEP 2 - ADD THE PRESSURE PLATE SWITCH

Slide Content: Bridge the black ground wire through the pressure plate switch. Look at the switch module and connect the wires carefully. The black wire goes from the buzzer, through the switch, and back to the battery. When some steps on the plate, the circuit will close!

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[BUZZER (Black leg)] ----> [SWITCH INLET] == [SWITCH OUTLET]  
                        ----> [BATTERY (Black)]
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 **SUPER ENGINEER! THE LOOP IS ALMOST COMPLETE!** 

SLIDE 9 — INTERACTIVE EXECUTION & TESTING

STEP 3 - TEST THE ALARM

Slide Content: Drop the virtual stone on the pressure plate to close the circuit. Watch carefully as the stone presses down on the mat. When the circuit closes, the electricity ants can finally complete their march all the way to the buzzer! Shabash! The alarm is ringing! BZZZZ!

SIMULATION SIMULATION: *"It's midnight in the orchard! Let's test our trap before the monkey arrives. Pick up that large virtual testing stone from your inventory panel and drop it right on top of our garden doormat switch. Look! The mat squishes down, the gap closes, the circuit completes, and list—BZZZZZZ! It works beautifully! Shabash!"*

SLIDE 10 — DEBUGGING & DIAGNOSTICS LEVEL 1

PUZZLE 1 - FIX THE WIRE POLARITY

Slide Content: Oops! The Wires Are Upside Down! Oh no! The buzzer isn't ringing because the wires are connected the wrong way. Can you swap the red and black wires to fix the circuit? Let's help the ants march the right way!

Debugging Concept - Directionality:

Electric current from direct-current toy elements flows in a specific direction. If the red and black supply polarities are completely inverted, the buzzer's internal logic block cannot process the path, keeping the alarm system silent.

PUZZLE 2 - REPLACE THE PLASTIC BLOCK

Slide Content: Oh no! There's a plastic block in the circuit and the buzzer won't ring! Plastic stops the electricity ants from marching. Find the metal pin to fix it! Spot the Problem! Hint: Conductive means ants can walk through!

Insulators (Plastic Blocks)

Materials like toys, plastic blocks, and rubber erasers completely block the electrical path. The ants hit a wall and cannot move forward.

Conductors (Metal Safety Pins)

Metals act like smooth, paved roads for electricity. Swapping out the plastic chunk for a metal pin opens up the highway for the ants.

PUZZLE 3 - SAFE OR NOT SAFE?

Slide Content: Which power source is safe for young inventors like you? Look carefully at each picture and choose wisely!

- Toy AA battery pack - Safe for kids!
- Wall socket - Never touch! Very dangerous!
- Battery with warning sign - Ask an adult first!

CRITICAL SAFETY NOTICE FOR CONTENT ARCHITECTS:

Ensure the contrast between a safe toy battery pack (1.5V to 9V direct current) and high-voltage alternate current domestic mains wall outlets (240V) is explicitly highlighted. Emphasize that household wall sockets are strictly dangerous.

STEM SAFETY SCIENTIST OATH

Slide Content: Wah! You are a safety superstar! I promise to only use toy batteries and never touch big wall plugs. I will always ask a grown-up before doing any electricity experiments!

INTERACTIVE RECITATION: *"Raise your right hand along with me, superstar inventors! Let's say our sacred safety pledge together: 'I promise to always use safe toy batteries for my inventions. I will never, ever touch big household wall plugs, and I will always ask my parents or teacher before starting an experiment!' Now you are an officially certified Planrs Academy Junior Lab Assistant!"*

FIND BUZZERS AT HOME & KEEP INVENTING!

Slide Content: Keep Exploring, Super Engineer! Doorbell: Press and hear the ding-dong sound! Fridge Alarm: Listen when the door stays open too long. School Bell: The buzzer that ends your classes! Thank You & Keep Inventing! The Midnight Mango Thief Adventure.

FINAL ENCOURAGEMENT: *"Ayaan's orchard is perfectly safe now, thanks to your incredible wiring skills! The sneaky monkey can't steal any more sweet fruits. Your homework today is to look around your kitchen and living room and spot one device that uses a buzzer. Write it down or draw it in your notebook to share next time. Thank you, and keep inventing!"*